Facial Recognition System on AWS (Serverless AI Architecture)

Facial Recognition System

Programme / Academy: Cloud Computing/ Cloudboosta

April 2026

Executive Summary

- Built a **fully serverless facial recognition system on AWS**

- Combines:
  - AI (Rekognition)
  - Serverless compute (Lambda)
  - API-driven architecture

- Achieves:
  - Real-time authentication (<1 sec)
  - High scalability
  - Secure identity verification

Business Case

- Organisation: **SecureVision Technologies Ltd**
- **Business Drivers:**
  - Improve security (eliminate impersonation)
  - Reduce operational cost (no infrastructure)
  - Enable real-time identity verification
  - Scale across multiple locations
- **Outcome:**
  - Pay-per-use model
  - Fully automated system
  - Enterprise-ready deployment

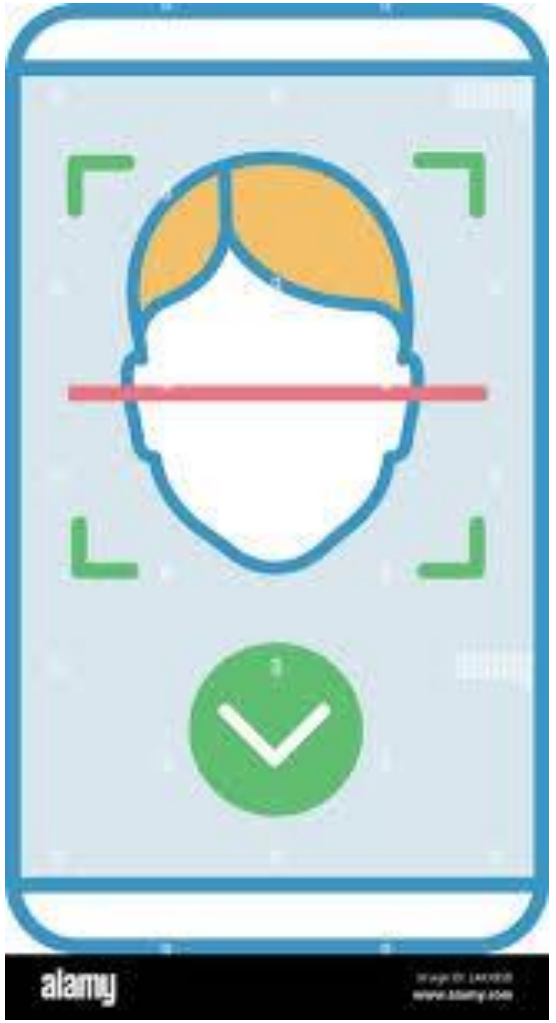
Problem Statement

- Traditional systems suffer from:
  - Identity theft (cards, PINs)
  - Poor scalability
  - Manual verification delays
  - High operational cost

- **Goal:**
  Design a **secure, scalable, automated identity system using AI + Cloud**

Project Objectives

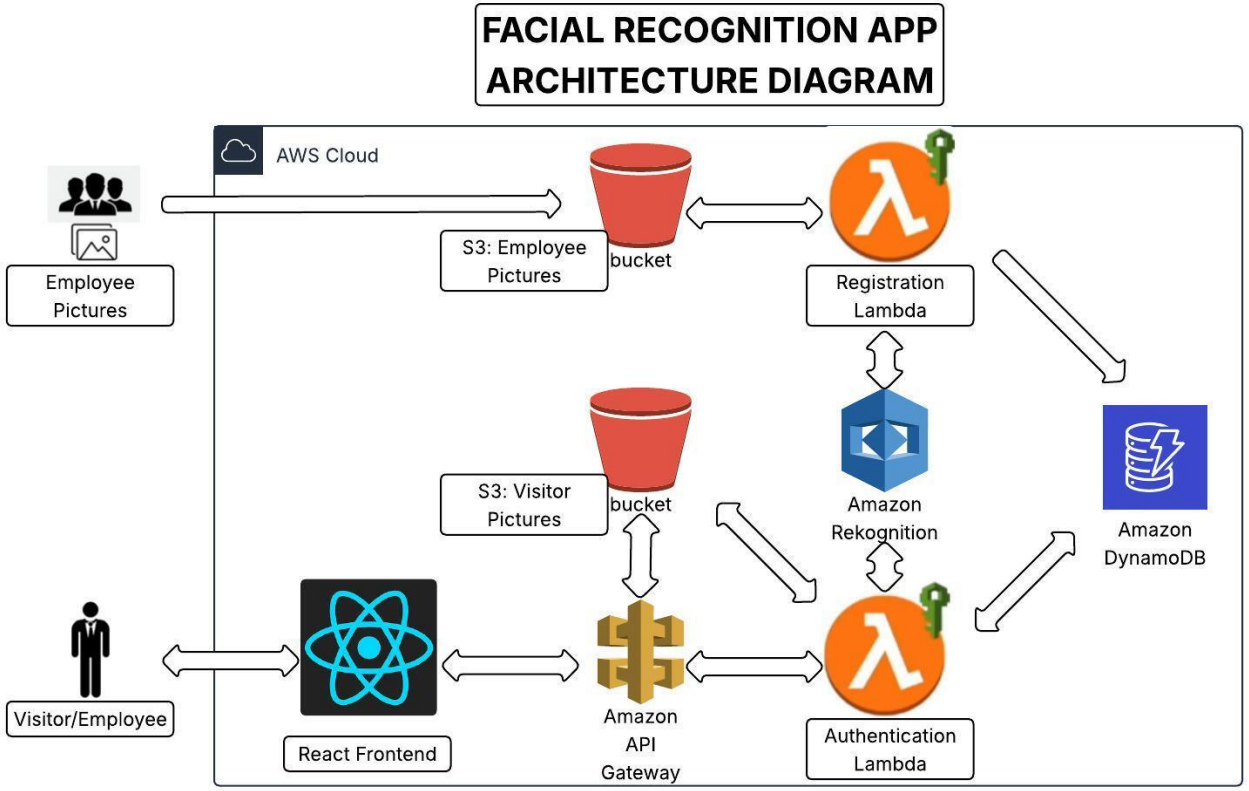
- Build a **serverless facial recognition system on AWS**
- Implement **registration using S3 and Lambda**
- Enable **authentication via API Gateway and Lambda**
- Integrate **Rekognition for face matching**
- Store data in **DynamoDB**
- Apply **IAM-based security and CloudWatch monitoring**
- **Outcome:**
  Scalable and secure real-time identity system



Solution Overview

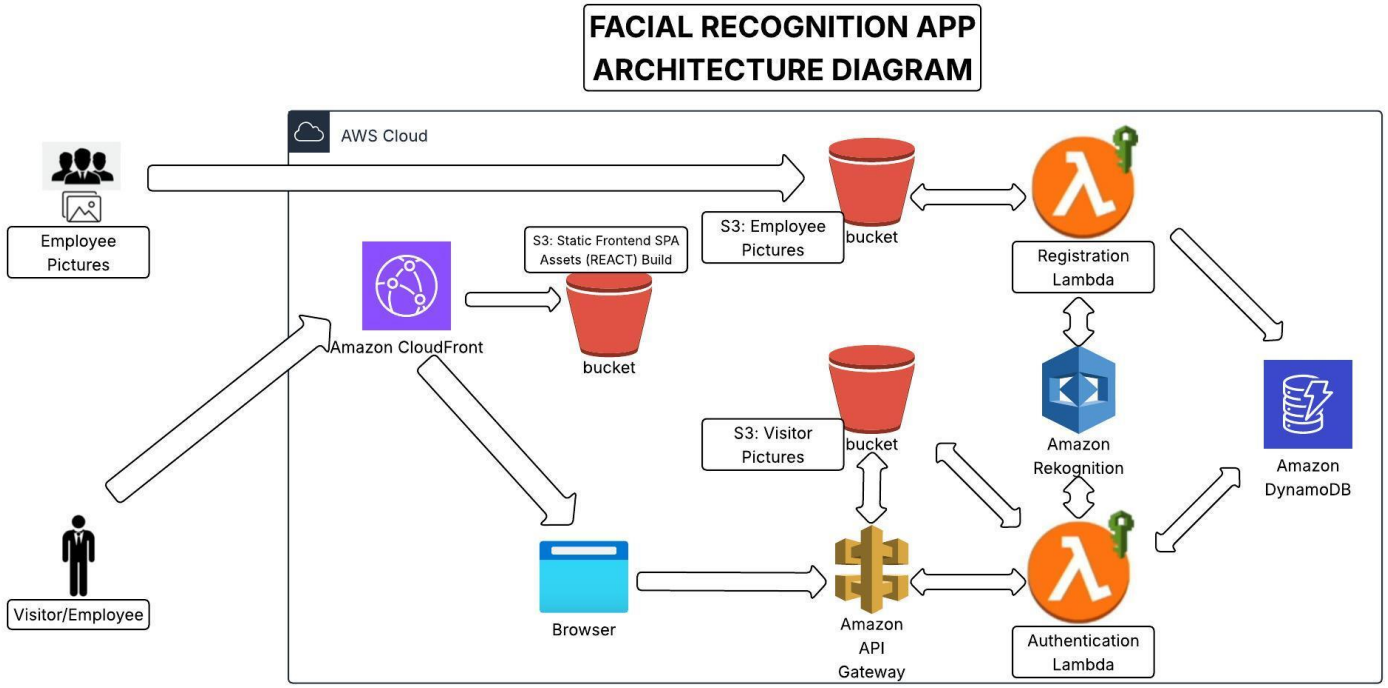
- Cloud-native facial recognition system
- Serverless + AI-powered
- Hybrid architecture:
  - Event-driven (registration)
  - API-driven (authentication)

- **Core Services:**
  - CloudFront, S3, Lambda, Rekognition, DynamoDB, API Gateway



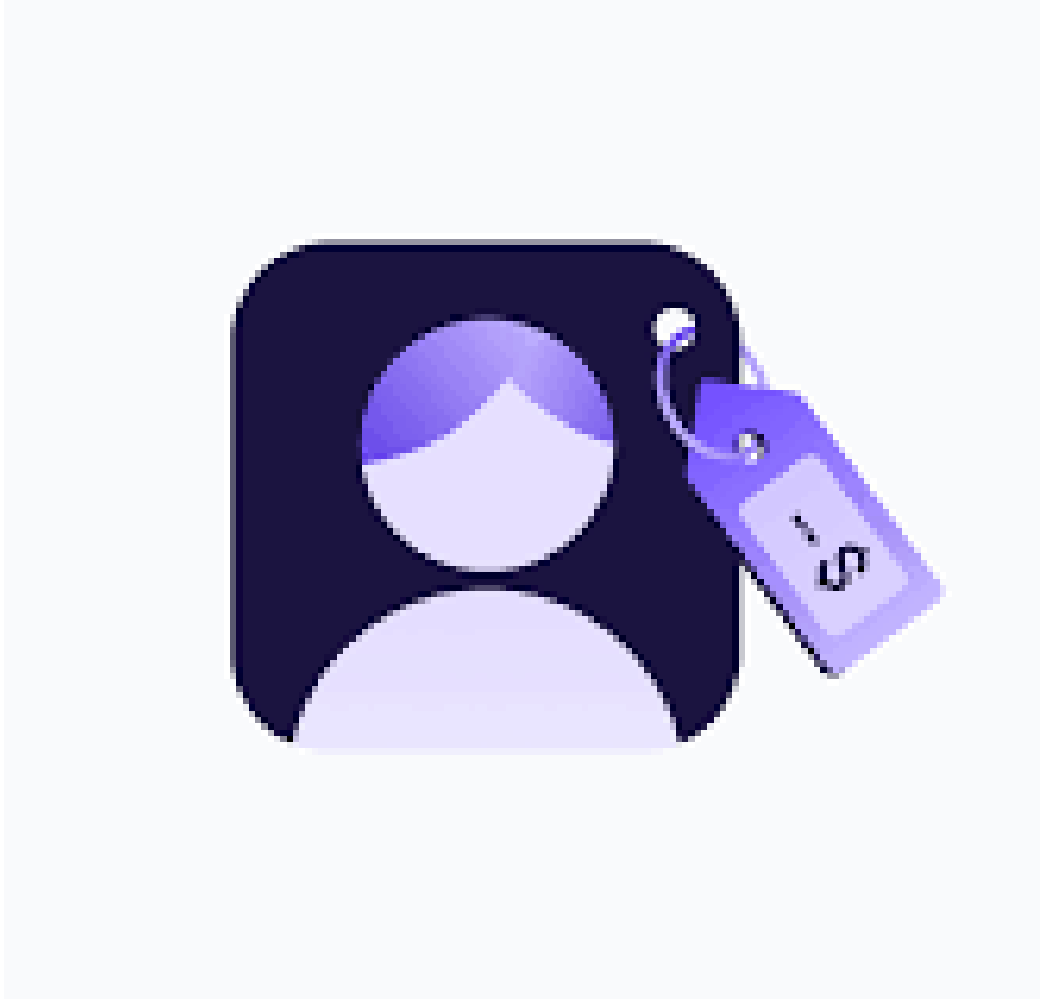
Architecture Overview

- **Layers:**
  - CloudFront
  - Frontend (React)
  - API Gateway
  - Lambda (Compute)
  - S3 (Storage)
  - Rekognition (AI)
  - DynamoDB (Database)
  - IAM (Security)
  - CloudWatch (Monitoring)



Project Artifacts & Demo

- Architecture diagram
  - https://drive.google.com/file/d/1E6xGUJlrZjnUrh0M8aAP4ml746IfN-D6/view?usp=drive_link
  - https://drive.google.com/file/d/1pzApoyZF39UlITK2mI1Dixmj8ncdTJfU/view?usp=drive_link
- Video recap of executed project
  - https://drive.google.com/file/d/1SutFomJvondX73Sy9ECcivWe6W4M0c8R/view?usp=drive_link



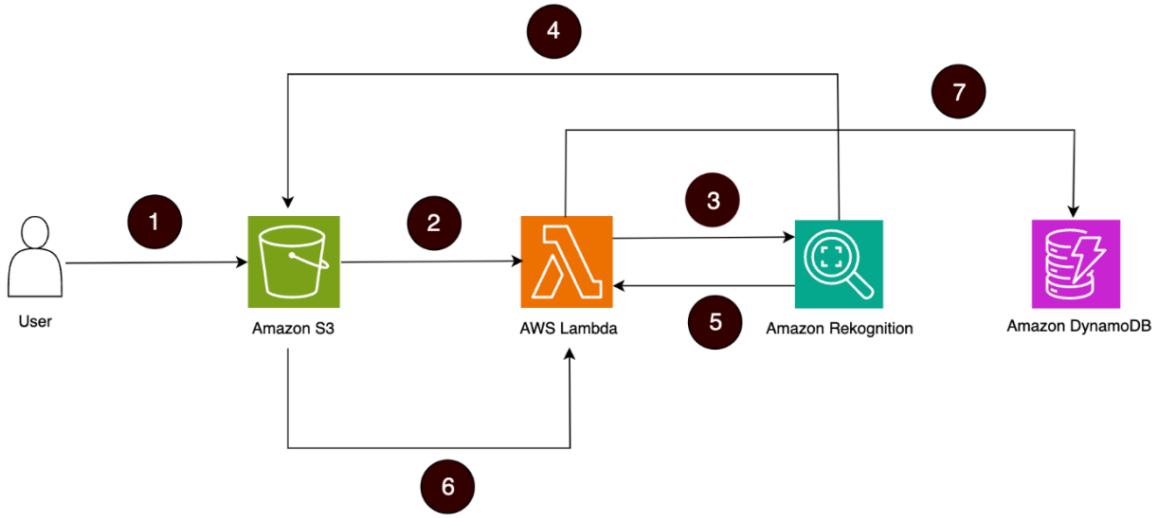
Architecture Style

- Fully serverless
- Event-driven + API-driven hybrid
- Decoupled microservices
- Stateless compute
- **Result:**
  - No infrastructure management
  - High availability
  - Elastic scaling

Registration Workflow (Event-Driven)

- User uploads image
- Stored in S3
- S3 triggers Lambda
- Rekognition indexes face
- Face ID stored in DynamoDB

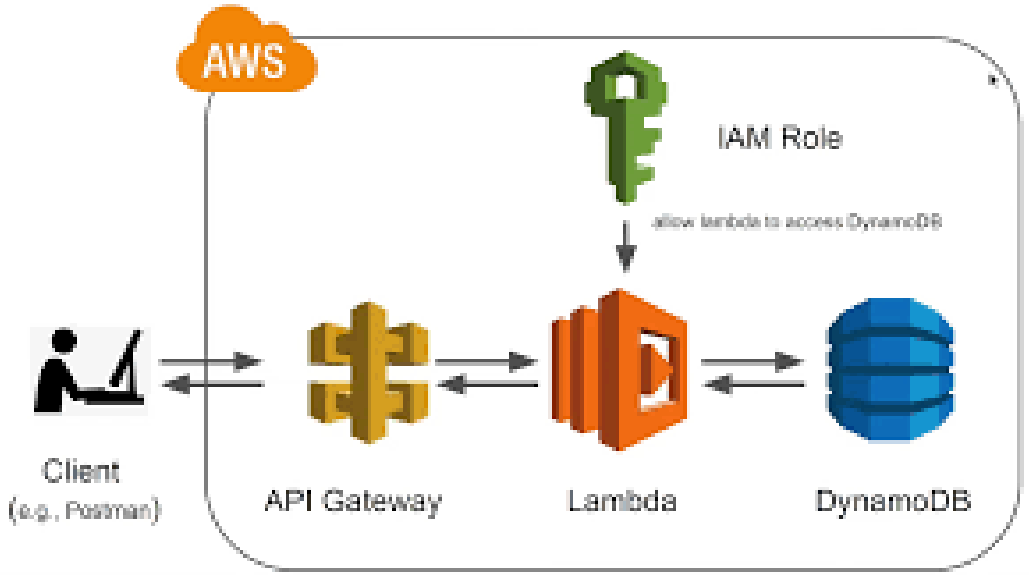
- **Key Insight:** Fully automated pipeline



Authentication Workflow (API-Driven)

- User uploads or captures image via browser (CloudFront-served app)
- Browser sends request to API Gateway
- Lambda processes image
- Rekognition compares face
- DynamoDB retrieves identity

- **Output:**
  - Match → Access granted
  - No match → Access denied



Technology Stack

AWS Lambda (Python 3.14)

Amazon Rekognition (AI engine)

Amazon S3 (storage)

DynamoDB (NoSQL DB)

API Gateway (REST API)

CloudFront-delivered (React Frontend)

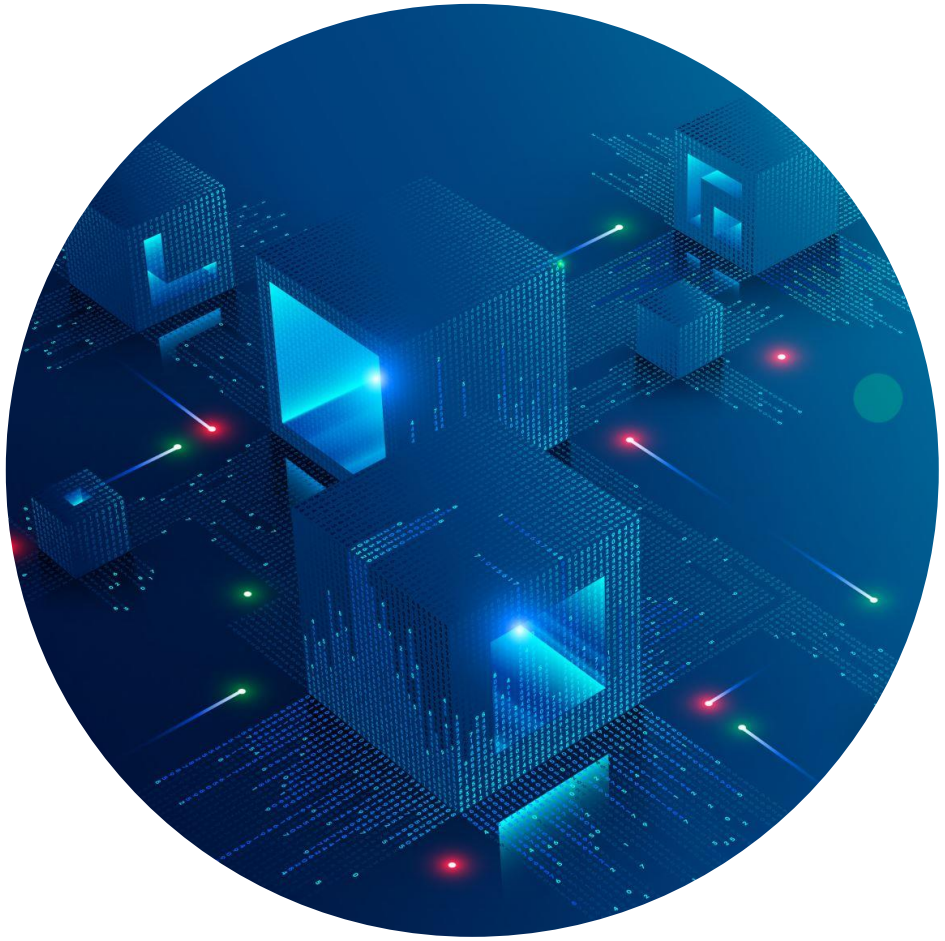
CloudWatch (monitoring)

IAM (security)

Key Design Principles

- Decoupling (independent services)
- Stateless compute (Lambda)
- Managed services (no servers)
- Event-driven automation

- **Impact:**
  - Scalability
  - Fault isolation
  - Simplicity



Security Architecture

- IAM role-based access control
- Least privilege principle
- S3 public access blocked
- API secured with keys
- Temporary credentials (STS)

- **Result:** Secure-by-design system

Performance & Scalability

- Lambda auto-scales
- DynamoDB on-demand scaling
- S3 unlimited storage
- Sub-second response time

- **Outcome:**
  - Handles high concurrency
  - Zero infrastructure limits

Observability & Monitoring

- CloudWatch logs:
  - Execution time
  - Errors
  - Memory usage

- Enables:
  - Debugging
  - Performance tuning
  - System auditing

Challenges Likely to be Encountered

- Image quality affecting recognition
- False positives/negatives
- IAM permission complexity
- API integration (CORS issues)
- Lambda cold starts

How Challenges Were Overcome

- Applied IAM roles for controlled access
- Enabled CORS in API Gateway
- Used CloudWatch for debugging
- Designed stateless architecture to reduce failure impact

- **Result:** Stable and production-ready system

Lessons Learned

- Serverless requires strong design discipline
- Event-driven architecture simplifies automation
- AI works well but needs calibration (accuracy, thresholds)
- Observability (CloudWatch) is essential for debugging
- Clean frontend–backend separation is critical
- IAM is central to system security
- Architecture thinking matters more than coding

AWS Well-Architected Pillars Applied

1. Operational Excellence

CloudWatch monitoring

Structured logging

2. Security

IAM roles

S3 access control

API security

3. Reliability

Serverless architecture

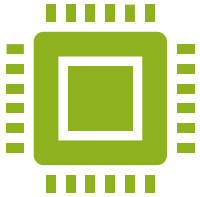
No single point of failure

AWS Well-Architected Pillars Applied

4. Performance Efficiency

Lambda auto-scaling

Rekognition optimised AI



5. Cost Optimisation

Pay-per-use model

No idle infrastructure

6. Sustainability

Serverless reduces resource waste

Business Value & ROI

- Reduced infrastructure cost
- Eliminated manual verification
- Improved operational efficiency
- Scalable enterprise deployment

- **Value Gains:**
  - Faster authentication
  - Lower operational overhead
  - Increased security compliance

Real-World Applications

- Corporate access control
- Banking (KYC)
- Healthcare identity systems
- Retail security
- Logistics verification

Results & Validation

- End-to-end system validated
- All services integrated successfully
- High accuracy recognition
- Reliable API responses

- **Evidence:**
  - CloudWatch logs
  - UI testing
  - DynamoDB records

Key Achievements

Fully serverless AI system

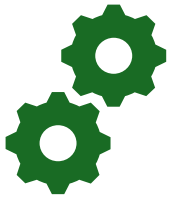
Real-time identity verification

Secure IAM-based architecture

Production-grade cloud solution

Conclusion

Successfully built a scalable, secure AI system



Demonstrated:

Serverless architecture

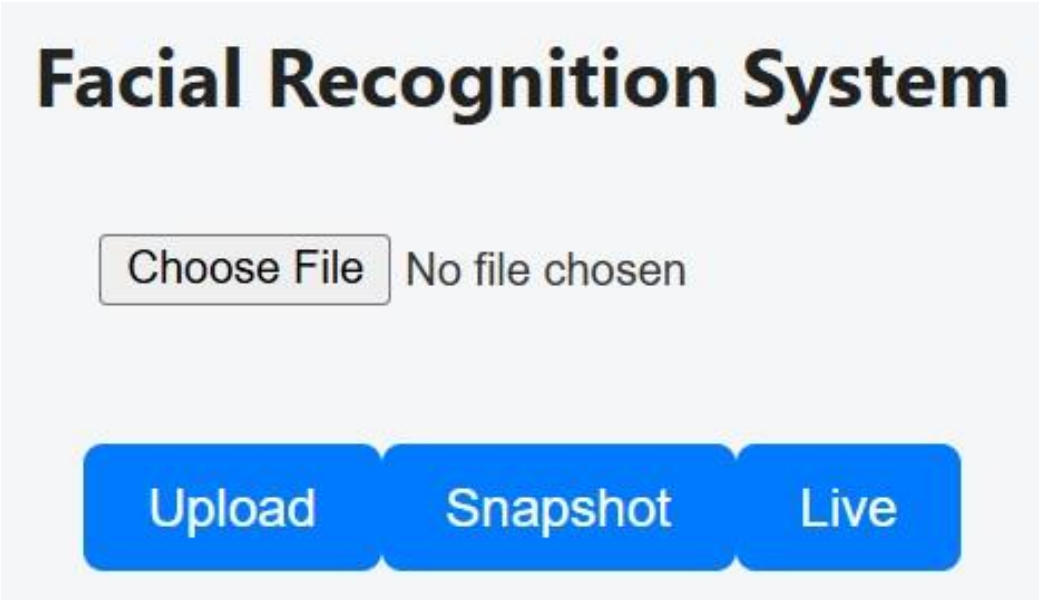
AI integration

DevOps practices

Final Insight: Cloud + AI enables intelligent, real-time systems at scale

Frontend Interface (User Input Layer)

d1msbxhxflfm9d.cloudfront.net



Displays the **user-facing interface** of the facial recognition system

Built using a **React-based frontend (Vite development server)**

Allows users to input **first name, last name, and upload an image**

Serves as the **entry point for authentication and registration workflows**

Sends user data and image to the backend via **API Gateway**

Provides a simple and intuitive interface for real-time interaction

Facial Recognition Match Results

Facial Recognition System

Status: Success

Time: 4/26/2026, 1:16:19 PM

Choose File jeff-bezos-2.jpg

Upload Snapshot Live

Employee Image

Visitor Image

Facial Recognition System

Status: Failure

Time: 4/26/2026, 1:18:03 PM

Choose File kevin-spacey-2.jpg

Upload Snapshot Live

Thank You